Punctuation Effects in Isolated Word Recognition

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Abstract

Current research in compound word processing has yielded mixed results in explaining the relationship between morphological decomposition and semantic transparency. In this study, I investigated the effects of decomposition by manipulating the presence of a hyphen across semantically transparent and semantically opaque compound words in a lexical decision task. Using a 2 x 2 within-subjects design, I found that semantically transparent and nonhyphenated words are recognised significantly faster than opaque and hyphenated words. However, there was no interaction found between these effects, suggesting that semantic transparency and forced morphological decomposition affect word recognition separately. I suggest that the divide between my study and previous literature may be a result of different materials, which may lead to insight on the creation of an appropriate model for compound word processing.

Punctuation Effects in Isolated Word Recognition

Many complex processes occur when a word is read. Research studying these processes mainly focuses on creating a cognitive model that can account for the varying factors that cause effects seen in visual word recognition. Popular models theorize the existence of a mental lexicon that holds representations of words, including their phonological, semantical and orthographical aspects, which influence the process of visual word recognition (Coltheart, 2006; Seidenberg, 2005). This lexicon is accessed every time a person reads. Most research has been conducted on recognition of single-morpheme words, which are words that have meaning and cannot be further broken down into meaningful constituents, such as 'snow'.

There has been some research exploring the visual recognition of words that are morphologically complex, such as 'snowed' that can be split into two separate units of meaning, 'snow' and 'ed'. It has been shown that word processing relies highly on morphemic structure, especially in early stages of visual processing (Rastle et al., 2000). Taft and Forster (1975) proposed the *morphological decomposition model*, as seen in Figure 1, which suggests that people process morphologically complex words by dividing them into their constituent morphemes and checking the lexicon for the stem word, such as 'snow' in 'snowed'.

Compound words are a type of morphologically complex word that combines two or more words. In other words, the morphological constituents of a compound word can individually carry meaning, such as 'snow' and 'ball' constitute the compound word 'snowball'. Taft and Forster (1976) refined their 1975 model to include compound words, proposing that instead of the prefix, the first morpheme of the word is searched for. This account assumes that words are read serially from left to right, and hence theorizes that compound words are broken

up with individual morphemes. However, other researchers have put forth the idea of a *whole word processing model* wherein there is no decomposition, and compound words are simply stored as any other word in the lexicon (Butterworth, 1983 as in Frisson et al., 2008). Researchers have since converged on the idea that each model is a possible route of word processing (Libben et al., 2020). A popular model by Pollatsek et al. (2000) proposed that it is not the case that one route is used over the other but that when a compound word is encountered, both routes are initiated in parallel in a *race*, and the route which will 'win' the race of processing the compound word depends on certain lexical characteristics of the word.

One of these lexical characteristics is semantic transparency, the extent to which the meaning of a compound word can be determined from the meaning of its morphemes. Words like 'snowball' are considered semantically transparent because the words 'snow' and 'ball' contribute to the meaning of 'snowball'; in contrast, 'butter' and 'fly' do not retain individual meaning in 'butterfly', making it a semantically opaque word. According to the race model (Pollatsek et al., 2000), the whole-word route is preferred for opaque words, whereas the decompositional route is faster for transparent words because it is assumed that it is easier to assemble the meaning of the transparent word from its constituents.

The effects of triggering the morphological decomposition route within the race model are often studied by visually splitting the morphemes. It has been shown that spatially separating the constituent morphemes of the compound word ('snowball' into 'snow ball') produces significant disruption to processing in lexical decision making and this disruption is more prominent for opaque words than transparent words (Experiment 4-6; Ji et al., 2011). These results support the race model theory that opaque words are accessed as whole words. However, other findings shed doubt on the race model: first, the interword space hinders processing of

transparent words too, which should not be the case if transparent words were decomposed in the first place, and second, there is no difference between transparent and opaque words at baseline without the space in both lexical decision (Experiment 3; Ji et al., 2011) and reading tasks (Experiment 1; Frisson et al., 2008).

However, the above papers study decomposition by introducing a space between two constituent morphemes, which may result in the morphemes being processed as separate words and contribute to the given results. They do not address the effect of visually separating morphemes while still linking them – that is, the effect of splitting a word into its constituent morphemes using an appropriate punctuation symbol, such as a hyphen. Häikiö et al. (2011) found that children read nonhyphenated words faster than hyphenated words within sentences and suggested that the presence of the hyphen cues decomposition similar to the interword space. – however, this evidence is complicated by two things. First, the study was conducted in Finnish, which is a more transparent language than English. Second, hyphenation effects were investigated at the sentence level, unlike the studies mentioned in the compound word processing literature which are based on results from a lexical decision task.

In the present study, I address the effects of hyphenation in cueing morphological decomposition in lexical decision tasks for both transparent and opaque English compound words in an adult population. I hypothesize, in line with Häikiö et al. (2011), that reaction time for hyphenated words will be longer than nonhyphenated words, as the presence of the hyphen will force decomposition and disrupt normal word processing. Further, extrapolating from the results of Jin et al. (2011), I predict that reaction time for opaque and transparent multi-morphemic constructions will be about the same. Lastly, I hypothesize that there will be an interaction effect: the semantic transparency effect will be more pronounced for hyphenated than

nonhyphenated words.

Methods

Participants

20 adults (13 women, 7 men, M_{age}=22.8 years, age range = 20-36 years) participated in this study via an online questionnaire. Demographic information of participants can be found in Table 1.

Materials

There were 200 stimuli in total, along with 20 additional stimuli shown for practice trials. Words rated on semantic transparency were taken from the Large Database of English Compound Words (LADEC; Gagné et al., 2019). The mean transparency rating of the LADEC words was found, and words 2 SD above the mean were considered transparent words and those 2 SD below mean were considered as opaque words. The word length, word frequency, orthographic neighbourhood size, orthographic Levenshtein distance (OLD), number of syllables, and number of morphemes for these words was generated using the English Lexicon Project database (Balota et al., 2007). 100 words matched on these characteristics and semantic transparency were obtained using Match (van Casteren & Davis, 2007), with 50 transparent words and 50 opaque words (see Table 2). 100 nonword distractors that matched the words in subsyllabic structure and transition frequencies were generated using the Wuggy algorithm (Keuleers & Brysbaert, 2010; see Table 3).

The stimuli were counterbalanced for hyphenation: 25 of the transparent words, 25 of the opaque words and their nonword counterparts were hyphenated for one half of the participants, whereas the other half of the participants were presented those words and nonwords as nonhyphenated and the rest of the words as hyphenated. In summary, every word and nonword

was hyphenated for at least half of the participants and no words were repeated across the hyphenated conditions.

Each participant was sent an online link to the experiment for data collection. The study could only be conducted on a device with an external keyboard, thus participants completed the study on their personal devices. The study was set-up to ensure that the stimuli were presented in in fullscreen mode, and all stimuli were displayed in white lowercase 20-point Courier font centred on a black background. Prior to the lexical decision task, participants were asked to input their demographic information (age, gender).

Design

A 2x2 fully-within design was used, with semantic transparency (transparent, opaque) and hyphenation (hyphenated, nonhyphenated) as independent variables. Each participant responded to words from four conditions (hyphenated-opaque, hyphenated-transparent, nonhyphenated-opaque and nonhyphenated-transparent) in the lexical decision task. The dependent variable was reaction time (RT) in milliseconds (ms).

Procedure

Participants were sent a link to the online study and asked to complete it within one sitting in a quiet place with minimal distractions. After inputting their age and gender, participants were told they will be shown a series of letter-string combinations one at a time. They were instructed to indicate whether they think the letter-string they are shown is a word by pressing the 'L' key or a non-word by pressing the 'A' key. Additionally, they were also instructed to respond irrespective of hyphenation – this was done by instructing them that both "baseball" and "base-ball" should be responded to as words. Lastly, participants were encouraged to answer as quickly and accurately as possible.

Participants first responded to 5 transparent words, 5 opaque words and 10 associated nonwords in a random sequence in a practice run to ensure familiarity with task requirements. This was followed by 200 test trials, in which participants responded to a random sequence of the 100 words and 100 nonwords. In each trial, participants first view a fixation point (+) in the middle of the screen for 500ms, indicating where the stimulus will appear. The screen would be blank for 100ms, then the stimulus would be presented either until the participant makes a response or 3000ms have passed without response. Participants were given feedback if they input an incorrect response to the letter string, with a message reading "Wrong answer, or too slow!" immediately after the incorrect response. There was no feedback given for correct responses.

Results

Responses were filtered using SPSS and statistical analyses were conducted using JASP. Only correct RTs for words were analysed. Out of 4000 total responses, responses that were incorrect, lapsed the 3000ms limit or were faster than 200ms were excluded, leaving a total of 3367 responses. Further, outliers were removed by excluding RTs that were ±2 SD from each participant's mean RT, leaving 3269 valid responses. Nonword responses were not analysed.

A two-way repeated-measures analysis of variance (ANOVA) revealed significant main effects of Hyphenation, $F(1, 19) = 37.47$, $M_{Se} = 45216.72$, $p < .001$, $\eta^2 = 0.331$. There was also a main effect of Semantic Transparency, $F(1, 19) = 15.98$, $M_{Se} = 17050.17$, $p < .001$, $\eta^2 = 0.125$, which supported what was predicted in Hypothesis 2. However, there was no interaction effect of Hyphenation x Semantic Transparency. The mean RTs are reported in Table 4 and plotted in Figure 2. A secondary analysis was conducted on accuracy rates, which revealed only a significant main effect of Semantic Transparency, $F(1, 19) = 47.45$, $M_{Se} = .26$, $p < .001$, $\eta^2 = 0.415$. The mean accuracy rates are reported in Table 5 and plotted by condition in Figure 3.

Discussion

This study investigated the effects of hyphenation and semantic transparency on word recognition in a lexical decision task. I found that hyphenated words are recognised significantly slower than nonhyphenated words, and semantically opaque compound words are recognised significantly slower than transparent compound words. However, there was no Hyphenation x Semantic Transparency interaction effect as I hypothesized – the presence of a hyphen did not differentially affect transparent and opaque words, nor did the transparency of a word differentially affect hyphenated and nonhyphenated words.

The observed hyphenation effect shows that a hyphen can be used as a forced morphological decomposition cue in English, similar to Finnish (Häikiö et al., 2011). Moreover, the hyphen might be a more appropriate cue for decomposition for lexical decisions, than a space. Compared to Jin et al. (2011), where the morphemes presented for lexical decision were split by a space, not a hyphen, the reaction times reported in this study (see Table 4) were faster, implying that interword spacing produces significantly more disruption to compound word processing. Given that much of the current literature of compound word lexical decisions use interword spaces, it may be important to consider that those latencies are inflated due to blank space separation instead of visually-linked separation methods as conducted in this study.

The observed semantic transparency effect shows that transparent words are recognized faster, and these effects are quite strong – they persist even when analysis on accuracy rates are taken into consideration, whereas hyphenation effects are not significant in this case (see Figure 3). The fact that this effect is seen despite no interaction effects is particularly interesting because it is inconsistent with current literature (Frisson et al., 2008; Ji et al., 2011): these studies found no difference in reaction times for opaque and transparent words in non-decomposed conditions,

suggesting that the race model (Pollatsek et al., 2000) was incorrect in assuming that opaque and transparent words are processed by different routes. However, this study's results imply that there is indeed a difference is the processing of semantic versus opaque words – but it challenges the race model because this difference is *not* due to decomposition by the hyphen. The mixed results of compound word LDTs have been explained by the different methods of operationalizing semantic transparency: it has been suggested that transparency effects can be attributed to semantic *compositionality* (Günther & Marelli, 2018), which is focuses on the generativity of a compound word's constituent morphemes. However, though this study did not use a compositionality measure, it used materials from the LADEC (Gagné et al., 2019) which takes into consideration the compositionality measure of semantic transparency, unlike the 1993 CELEX database used in both Frisson et al. (2008) and Ji et al. (2011).

Further research in the field of morphological decomposition and compound word processing would hence benefit from a replication of existing studies using the CELEX database with the LADEC materials instead. Such research might also further investigate the nature of decomposition cues by comparing interword spacing and hyphenation within the same study. These studies may be smaller steps towards solving the bigger question of transparency in compound word processing, which is the creation of a lexical model that can sustain these observed effects and sufficiently explain their existence.

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Appendix

Table 1. Demographic details of participants.

	n	M_{age}	SD	Minimum	Maximum
Male	7	24.27	5.25	21	36
Female	13	22.00	1.35	20	25
Total	20	23.14	3.31	—	—

Table 2. List of words used in the experiment by semantic transparency.

Semantically Transparent words		Semantically Opaque words	
daydream	necktie	keynote	highbrow
schoolmate	spillover	rosewood	blackball
footprint	beehive	floodlight	blueprint
pawnshop	workday	egghead	ponytail
barmaid	surfboard	wayside	taproom
bathrobe	airflow	cloudburst	whiplash
washbowl	bathroom	sourpuss	werewolf
lighthouse	mailbag	drawbridge	playwright
teakettle	dustpan	freedman	topknot
graveyard	footstep	carpetbag	warfare
steamship	flashbulb	rearguard	sunfish
doorway	handrail	hothead	hearsay
handgun	torchlight	breakfast	firebug
blackbird	spotlight	bedrock	bulldog
knockdown	meatball	beachhead	chestnut
lawman	weekday	wallflower	backdrop
topsoil	sheepskin	watershed	crackdown
grapevine	hairpin	knothole	lineman
hideaway	downgrade	pothole	sagebrush
guesthouse	bookmark	feedback	grandstand
eyeball	yardstick	casebook	lifeline
wavelength	watchman	mainspring	jukebox
wineglass	soybean	gunboat	bedpan
cutoff	workhouse	dateline	boardroom

wildfire	flaxseed	broadcast	bellboy

Table 3. List of words used in the experiment by semantic transparency.

Semantically Transparent words		Semantically Opaque words	
swimlash	tadloom	weapday	steedship
speedlight	genboot	beakmave	daregrade
flueglint	cloadman	squealmate	foodspint
backchap	heatpay	beehert	fermaway
potrile	carpetsob	wornnay	flakreed
werecelf	elfhood	flashcurl	doldream
calfrook	wagsill	wifenire	baredobe
playwrects	bourtroom	waselentes	tightnouse
notersted	tolblot	worthound	ashplow
brookcast	broadfast	clerkover	deedway
histcrow	monfish	withdrass	maintag
warwire	fallbug	libman	nailtie
bendsoy	domeline	baseboom	rushpan
boosthead	ploudbulbs	hairlon	euebale
sundbrush	knotmile	spackbind	grasewine
hutbead	maststring	yardstand	palkship
fistman	toadback	teawontle	shourskin
chessrut	jolkbox	hashraid	tomsoid
popy-told	keygose	barmeon	wrockcown
blandstack	brackdogs	spidlight	gudgehound
bellcog	betdock	washbeon	gainsman
drawflipes	flackbake	tirthlight	moodball
rearbudge	romehood	soymien	cudoil
budran	watecromer	foottrap	sawnboard

sourciss	filkline	gralmyons	hashrun

Table 4. Mean, SD and SE of RTs by condition

Condition	n	M_{RT}	SD	SE
Transparent-Hyphenated	20	736.105	131.720	29.453
Transparent-Nonhyphenated	20	692.367	121.203	27.102
Opaque-Hyphenated	20	769.113	132.842	29.704
Opaque-Nonhyphenated	20	717.755	110.389	24.684

Table 5. Mean, SD and SE of accuracy rates by condition

Condition	n	$M_{accuracy}$	SD	SE
Transparent-Hyphenated	20	0.904	0.081	0.018
Transparent-Nonhyphenated	20	0.788	0.136	0.030
Opaque-Hyphenated	20	0.910	0.073	0.016
Opaque-Nonhyphenated	20	0.798	0.113	0.025

Figure 1. The morphological decomposition model by Taft and Forster (1975).

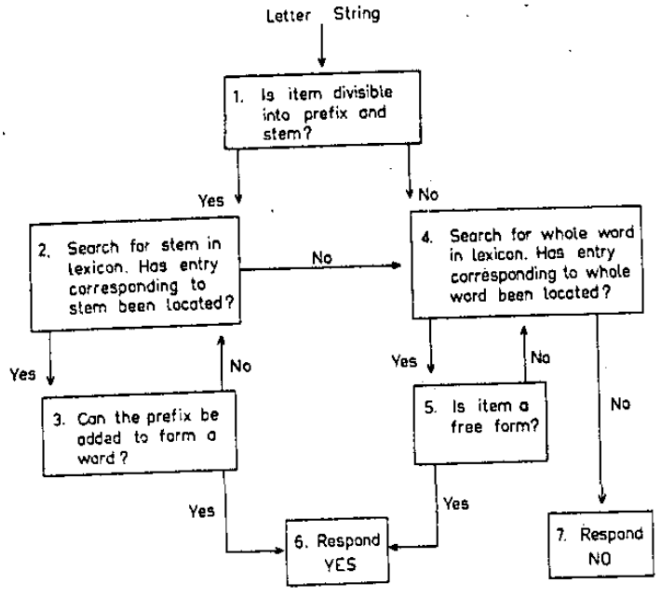


Figure 2. Results of repeated measures ANOVA on mean RTs.

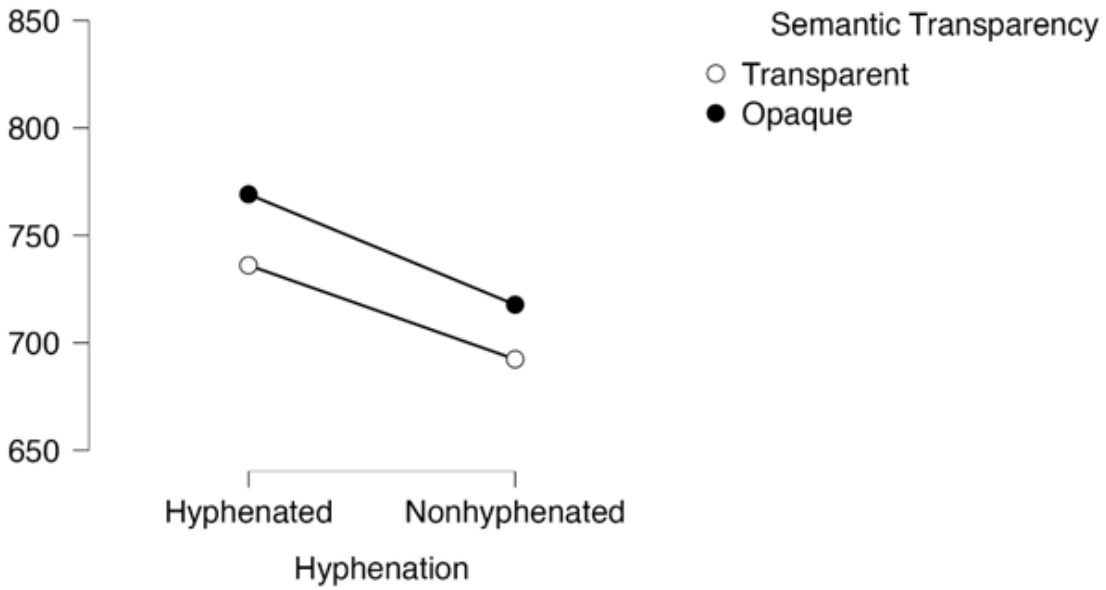


Figure 3. Results of repeated measures ANOVA on accuracy.

